



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

MOOC Student
August



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodologies

- Data Collection via Web scraping
- Data Wrangling
- Complete the EDA with Visualization
- Dashboard

All results

- Predict first stage of the Falcon 9 lands successfully

Introduction

Project background and context

- Space Y wants to launch rockets with minimal costs

Problems you want to find answers

- Determine the price of each launch
- Determine if SpaceX will reuse the first stage

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Using SpaceX API and Web scraping
- Perform data wrangling
 - Use flowcharts and key phrases
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Built, evaluated, improved, and found the best classification model

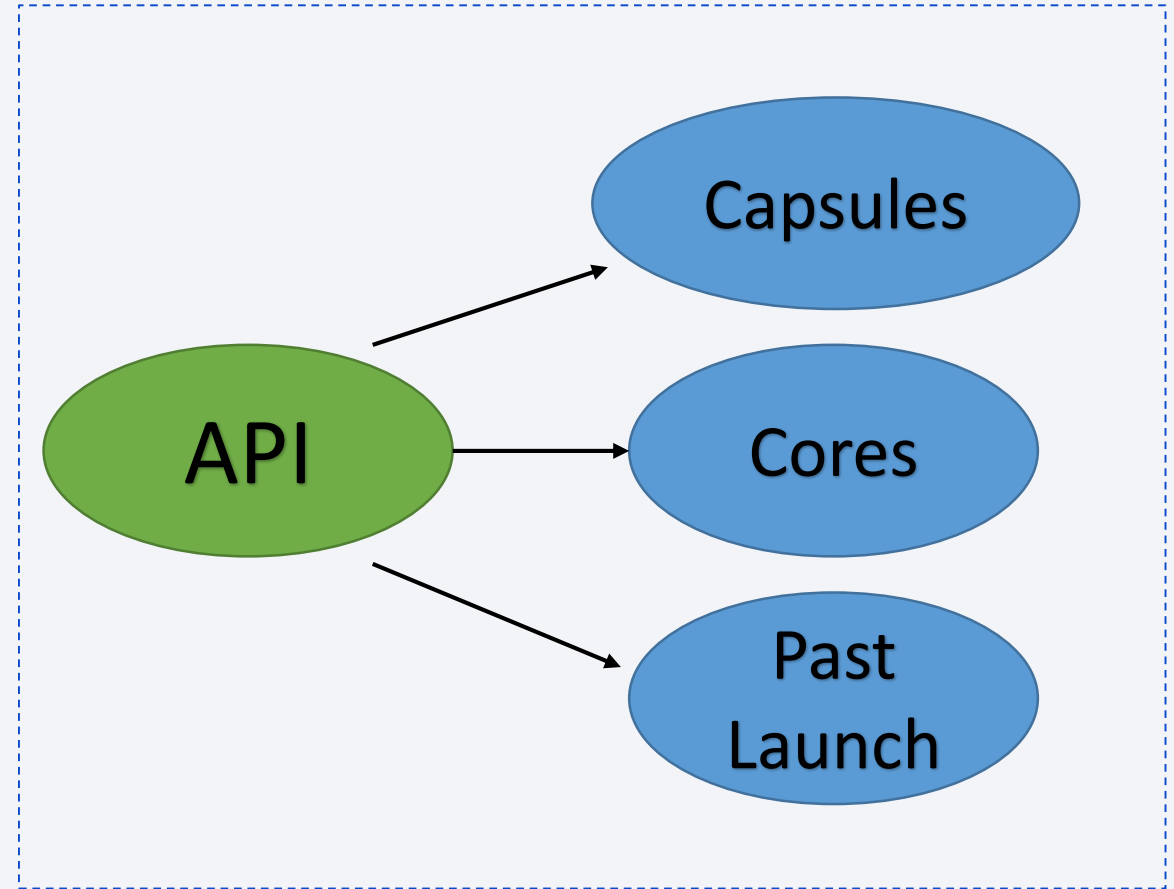
Data Collection

Describe how data sets were collected

- API will give us data about launches, including information about the rocket used, payload delivered, launch specifications, landing specifications, and landing outcome
- Web scrape some HTML tables that contain valuable Falcon 9 launch records

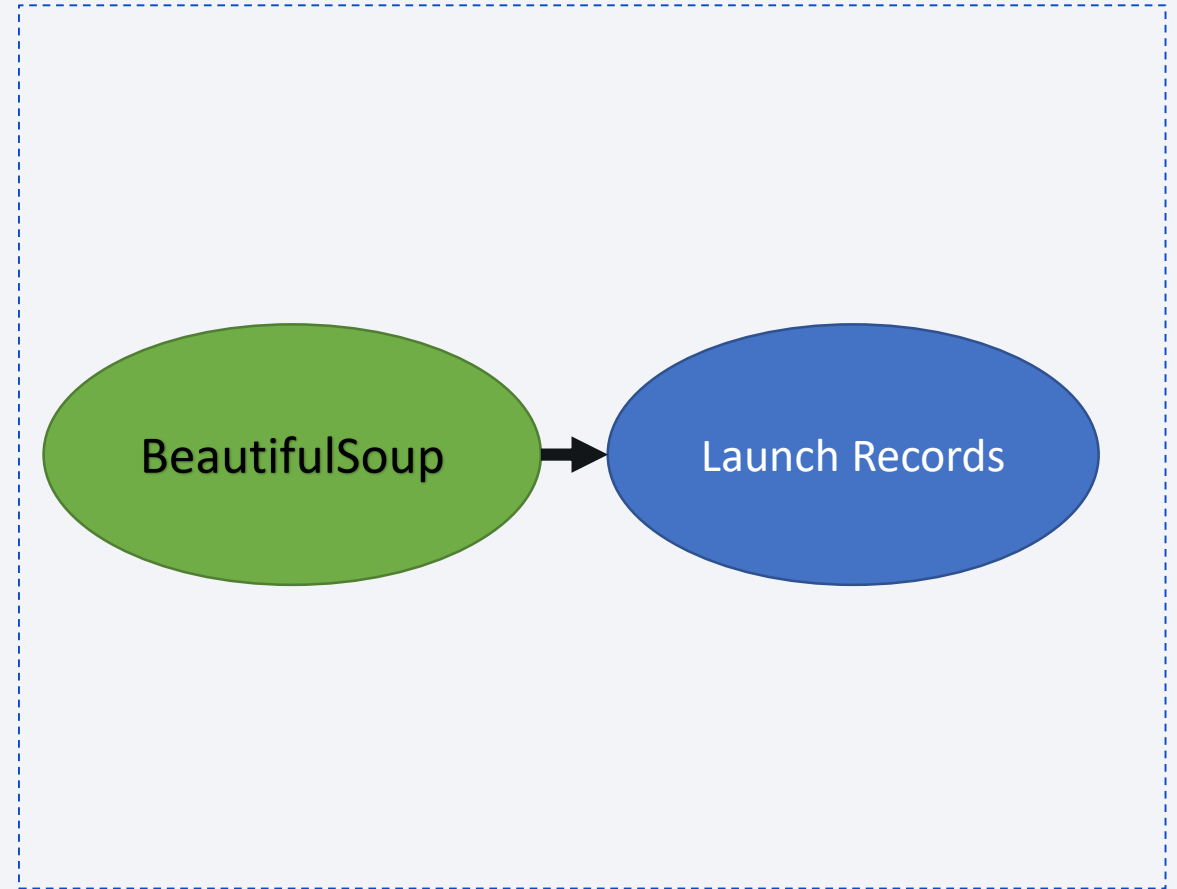
Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- GitHub URL of the completed SpaceX API calls notebook
- <https://github.com/dennislamcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- GitHub URL of the completed web scraping notebook,
- <https://github.com/dennislmcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/jupyter-labs-webscraping.ipynb>



Data Wrangling

Describe how data were processed

- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Calculate the number and occurrence of mission outcome per orbit type
- Create a landing outcome label from outcome column
- You need to present your data wrangling process using key phrases and flowcharts
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose
- <https://github.com/dennislamcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

Summarize what charts were plotted and why you used those charts

- Visualize the relationship between Flight Number and Launch Site
- Visualize the relationship between Payload and Launch Site
- Visualize the relationship between success rate of each orbit type
- Visualize the relationship between FlightNumber and Orbit type
- Visualize the relationship between Payload and Orbit type
- Visualize the launch success yearly trend
- <https://github.com/dennislamcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/jupyter-labs-eda-dataviz.ipynb>

EDA with SQL

Summarize the SQL queries you performed

- Retrieve the most recent date from the SpaceX table
- Display the minimum payload mass
- Total payload_mass_kg carried by the booster versions
- Display 5 records launched on Friday
- Unique launch sites
- <https://github.com/dennislamcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/jupyter-labs-eda-sql-coursera.ipynb>

Build an Interactive Map with Folium

Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map

- Mark all launch sites on a map
- Mark the success/failed launches for each site
- Calculate the distances between a launch site to its proximities

Explain why you added those objects

- https://github.com/dennislamcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Summarize what plots/graphs and interactions you have added to a dashboard

- Analyzing launch site geo and proximities
- Choose an optimal launch site

Explain why you added those plots and interactions

<https://github.com/dennislamcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/Build%20an%20Interactive%20Dashboard%20with%20Ploty%20Dash.ipynb>

Predictive Analysis (Classification)

Summarize how you built, evaluated, improved, and found the best performing classification model

Create 2 classes

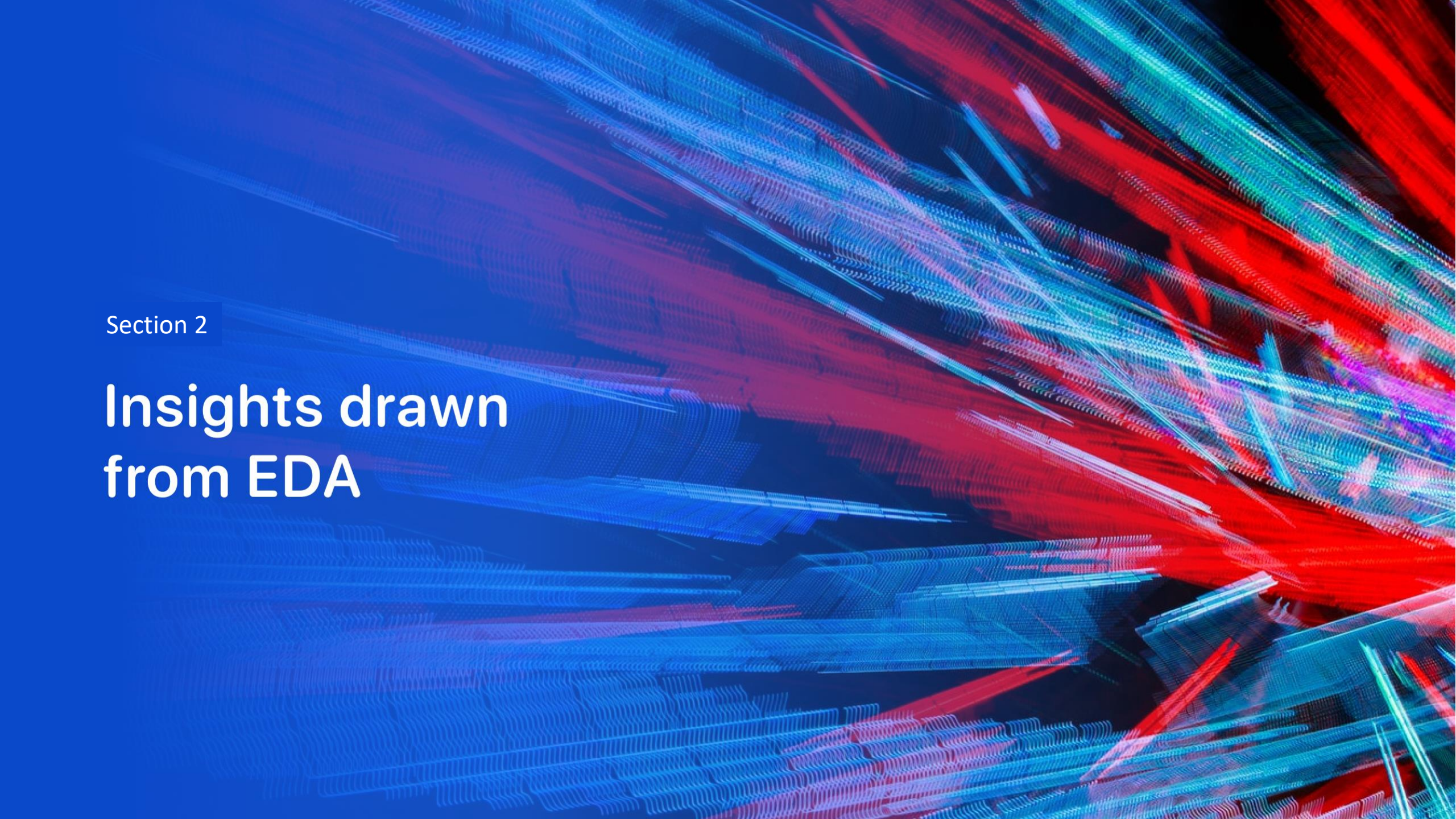
Standardize the data, create a logistic regression object then create a GridSearchCV.

You need present your model development process using key phrases and flowchart

https://github.com/dennislamcv1/IBMDS/blob/main/Applied%20Data%20Science%20Capstone%202022/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

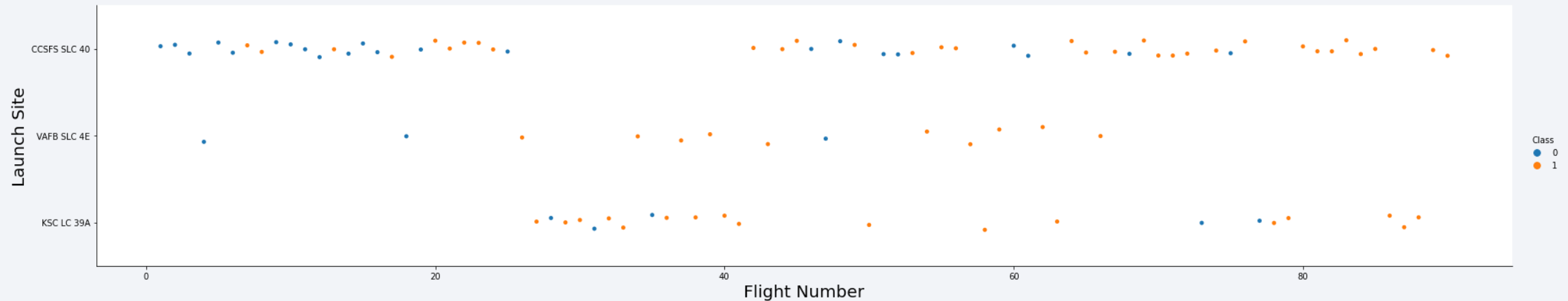
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

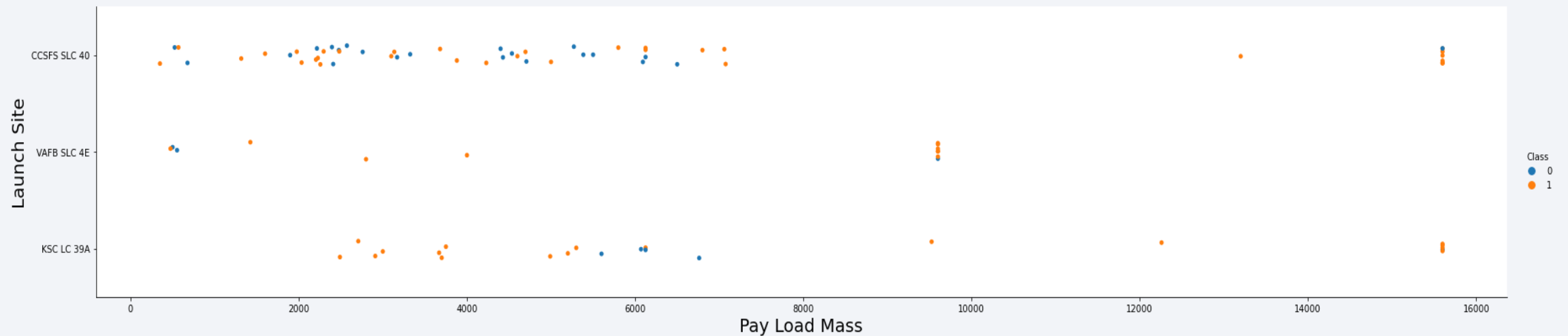
Insights drawn from EDA



1. *Journal of Management Studies*, 1996, 33, 1, 1-14.

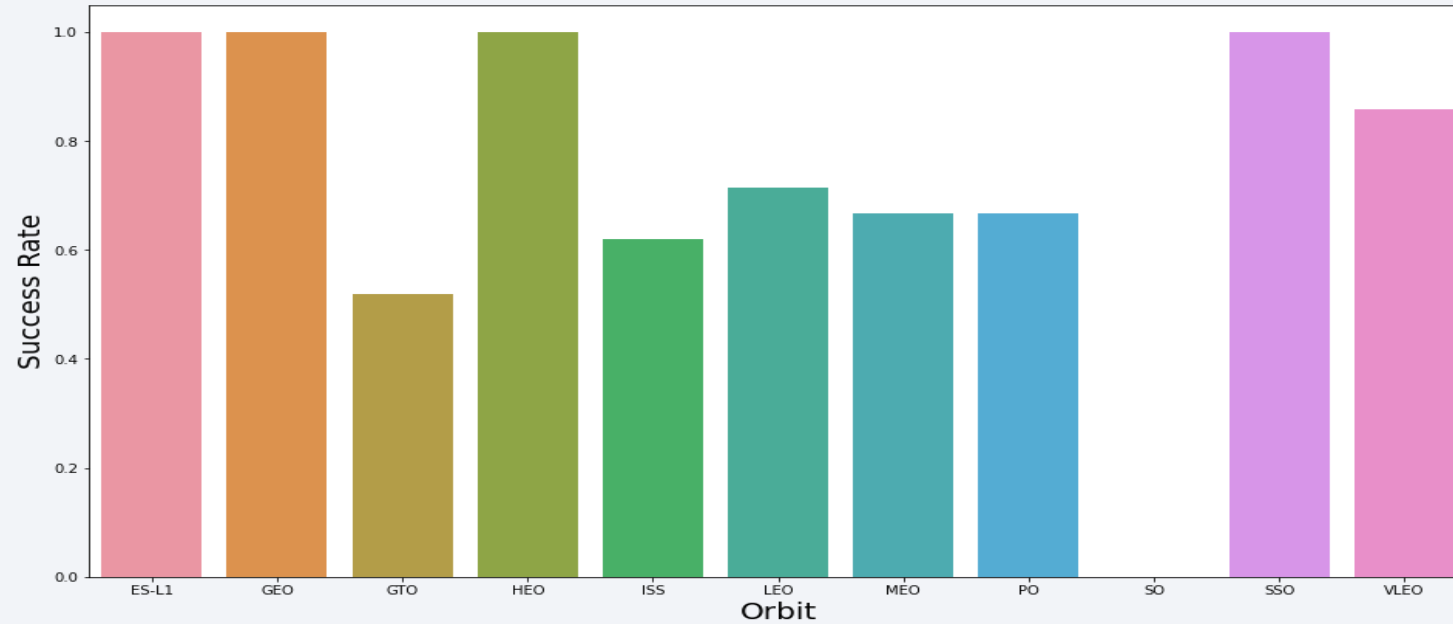
Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site



- Majority less than 700 Pay Loads

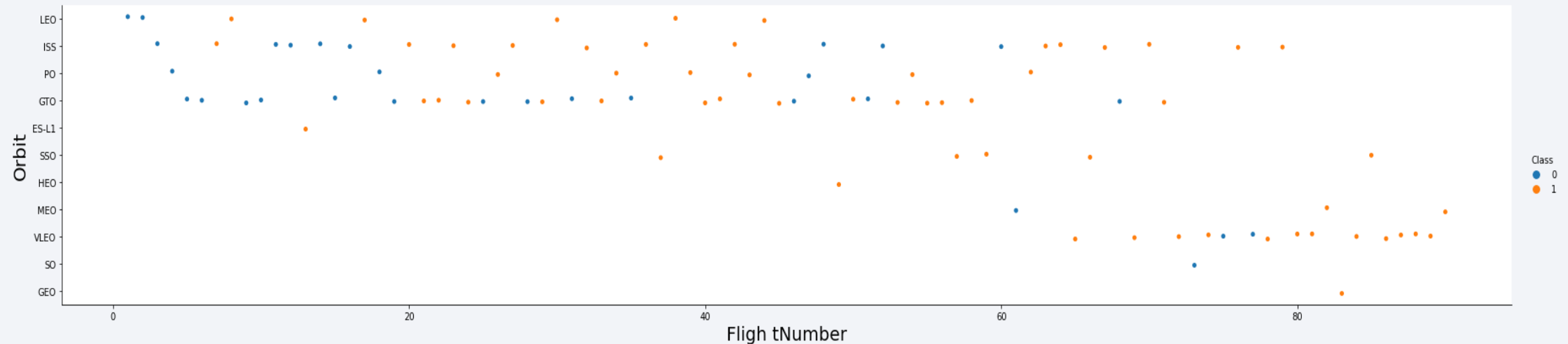
Success Rate vs. Orbit Type



There are 4 equally orbit launched

Flight Number vs. Orbit Type

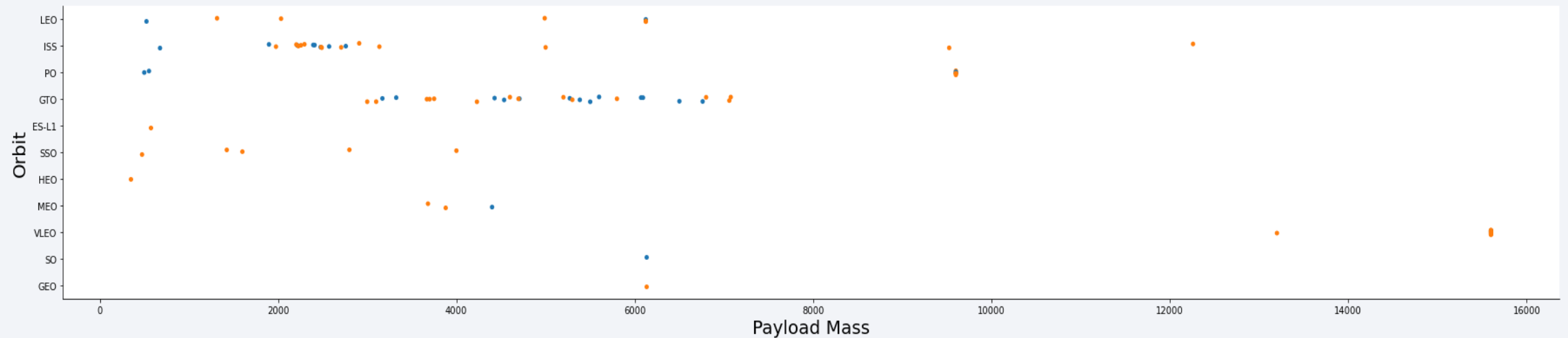
- Show a scatter point of Flight number vs. Orbit type



- Class 0 and 1 evenly

Payload vs. Orbit Type

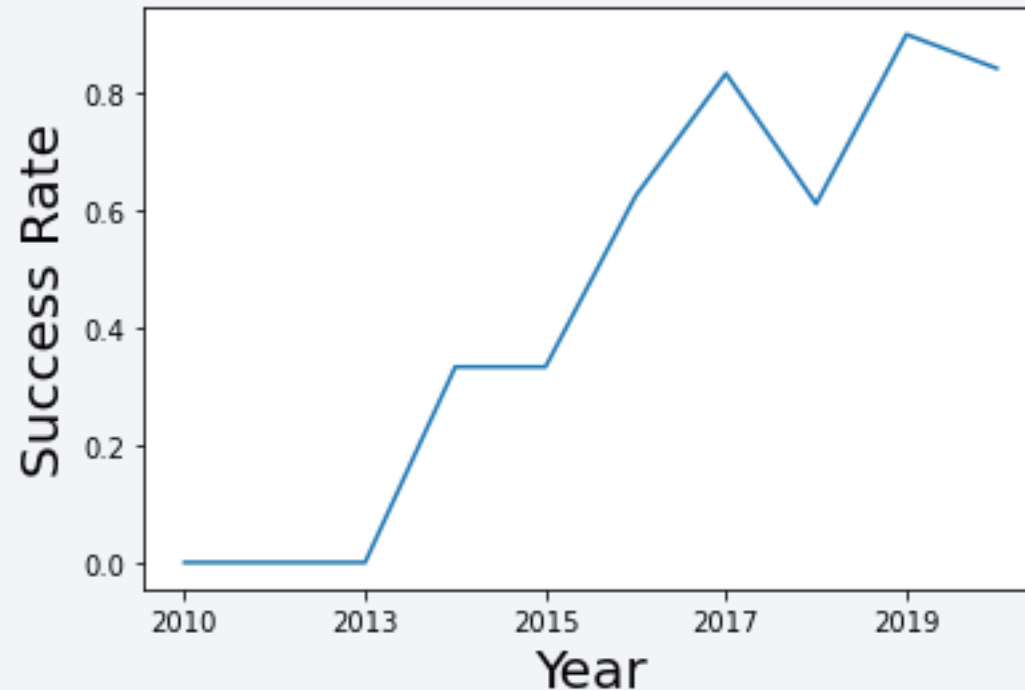
- Show a scatter point of payload vs. orbit type



- Payload less than 7000

Launch Success Yearly Trend

- Show a line chart of yearly average success rate



- Each year improved success rate

All Launch Site Names

- Find the names of the unique launch sites
- Present your query result with a short explanation here

Task 1

Display the names of the unique launch sites in the space mission

```
pd.read_sql_query("SELECT DISTINCT Launch_Site FROM SPACEX \n                    LIMIT 5;", db)
```

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Present your query result with a short explanation here

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
pd.read_sql_query("SELECT * FROM SPACEX \n                    WHERE Launch_Site LIKE 'CCA%' \n                    LIMIT 5;", db)
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_
0	04-06-2010	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	C
					Dragon demo	

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Present your query result with a short explanation here

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS) [↑](#)

```
pd.read_sql_query("SELECT SUM (PAYLOAD_MASS_KG_) FROM SPACEX \n\nWHERE Customer LIKE '%NASA (CRS)%';", db)
```

SUM (PAYLOAD_MASS_KG_)	
0	48213

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

Task 4

Display average payload mass carried by booster version F9 v1.1

```
pd.read_sql_query("SELECT AVG (PAYLOAD_MASS_KG_) FROM SPACEX \n                   WHERE Booster_Version = 'F9 v1.1';", db)
```

	AVG (PAYLOAD_MASS_KG_)
--	------------------------

0	2928.4
---	--------

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here

```
df[df["Landing_Outcome"] == "Success (ground pad)"].head(3)
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYL
19	22-12-2015	01:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm- OG2 satellites	
26	18-07-2016	04:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	
29	19-02-2017	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- Present your query result with a short explanation here

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
df2 = df[df["Landing_Outcome"] == "Success (drone ship)"]  
df2
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mis
22	08-04-2016	20:43:00	F9 FT B1021.1	CCAFS LC-40	SpaceX CRS-8	3136	LEO (ISS)	NASA (CRS)	
23	06-05-2016	05:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group	

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here

Task 7

List the total number of successful and failure mission outcomes

```
df["Mission_Outcome"].value_counts()
```

```
Success          98
Failure (in flight)  1
Success (payload status unclear)  1
Success          1
Name: Mission_Outcome, dtype: int64
```

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here

```
df3 = df[(df["Landing_Outcome"] == "Failure (drone ship)") & (df["Year"] == 2015)]
df3
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Custom
13	2015-10-01	09:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NA (CF
16	2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NA (CF

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Present your query result with a short explanation here

```
df3 = df[(df["Landing_Outcome"] == "Failure (drone ship)") & (df["Year"] == 2015)]  
df3
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Custom
13	2015-10-01	09:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NA (CF
16	2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NA (CF

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) descending order

```
df4 = df[df["Date"].between("2010-04-06", "2017-03-20")]
```

```
df4["Landing _Outcome"].value_counts()
```

No attempt	10
Failure (drone ship)	5
Success (ground pad)	5
Success (drone ship)	5
Controlled (ground)	2

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

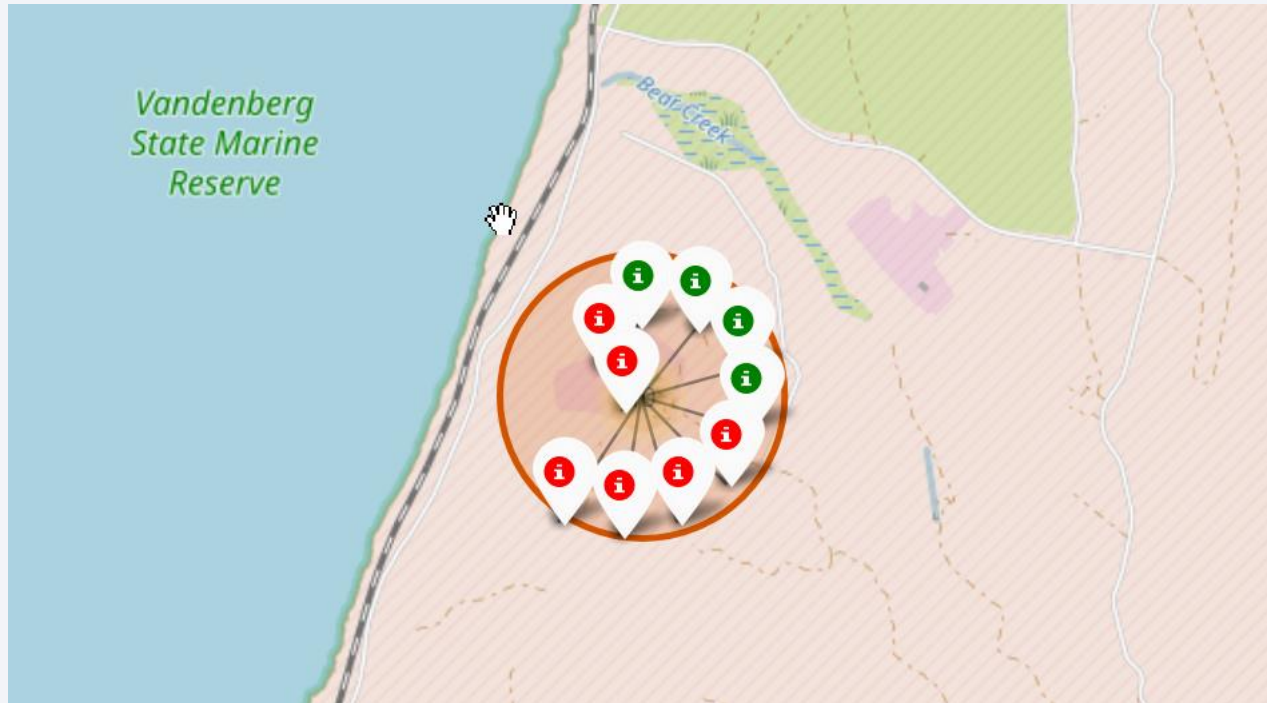
Launch sites on a map

Explain the important elements and findings on the screenshot



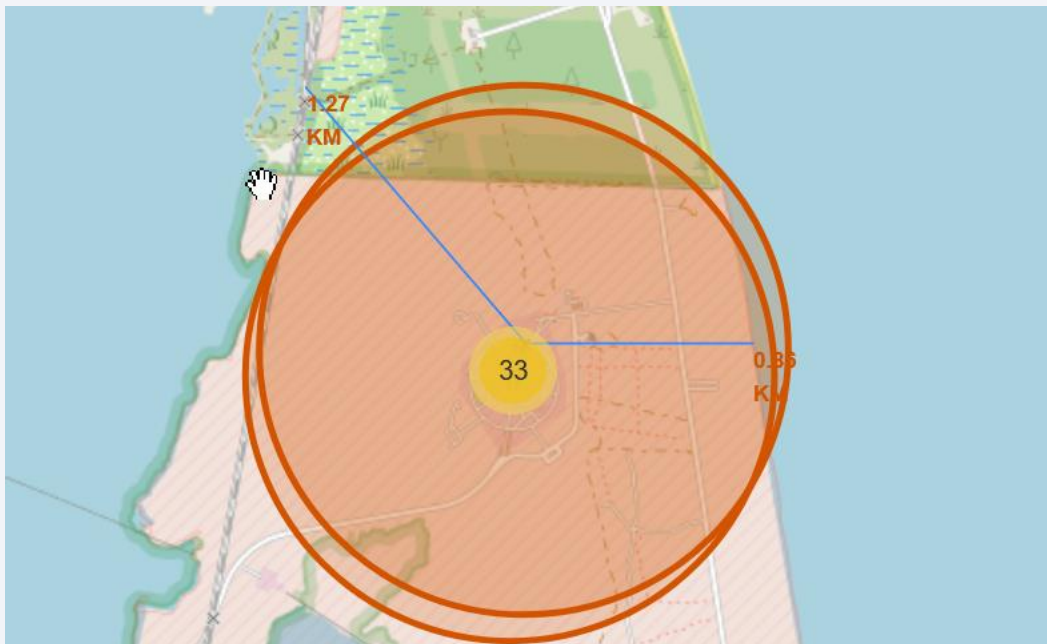
Success/failed launches for each site

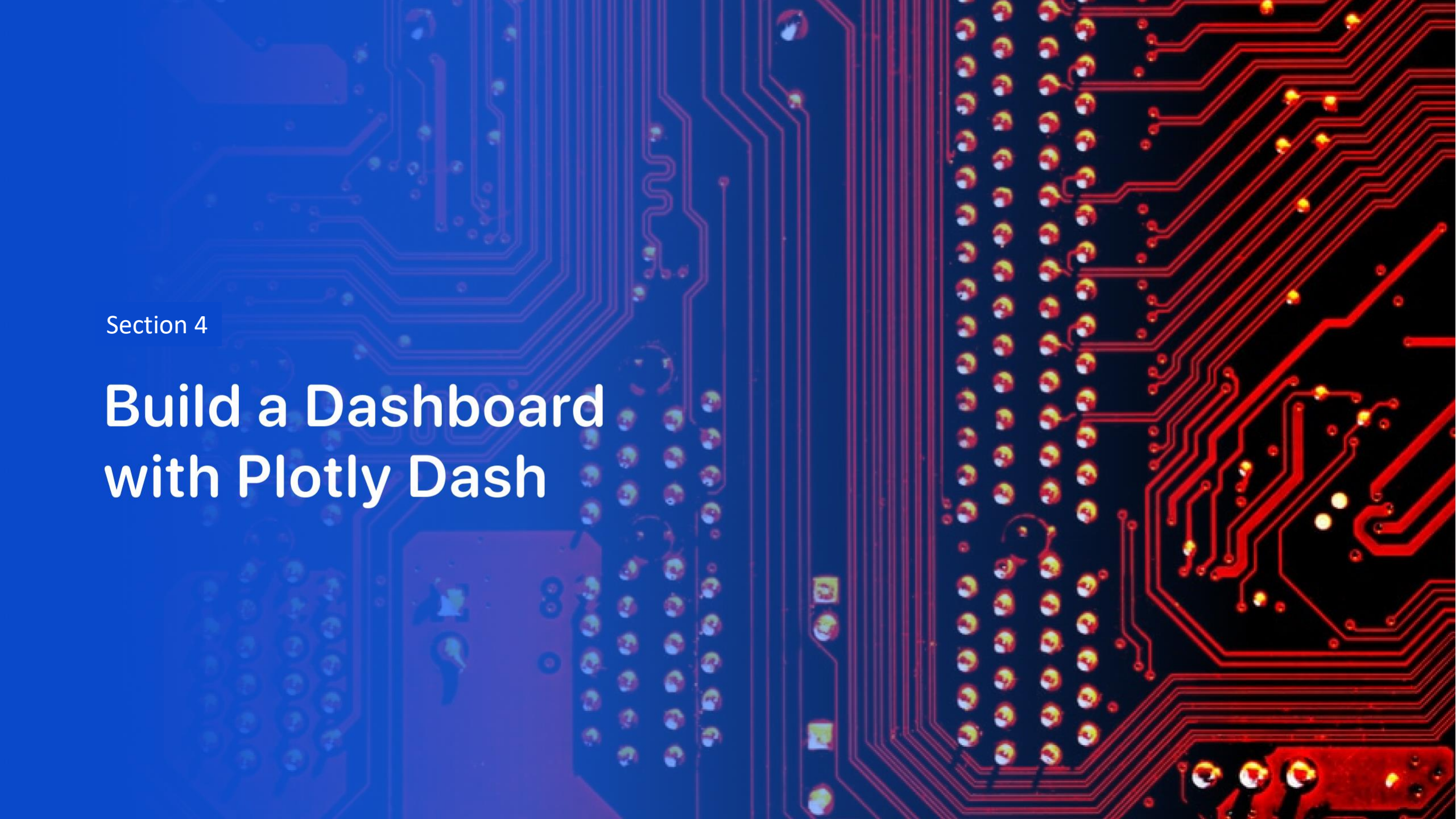
- Explain the important elements and findings on the screenshot



Distances between a launch site to its proximities

- Explain the important elements and findings on the screenshot



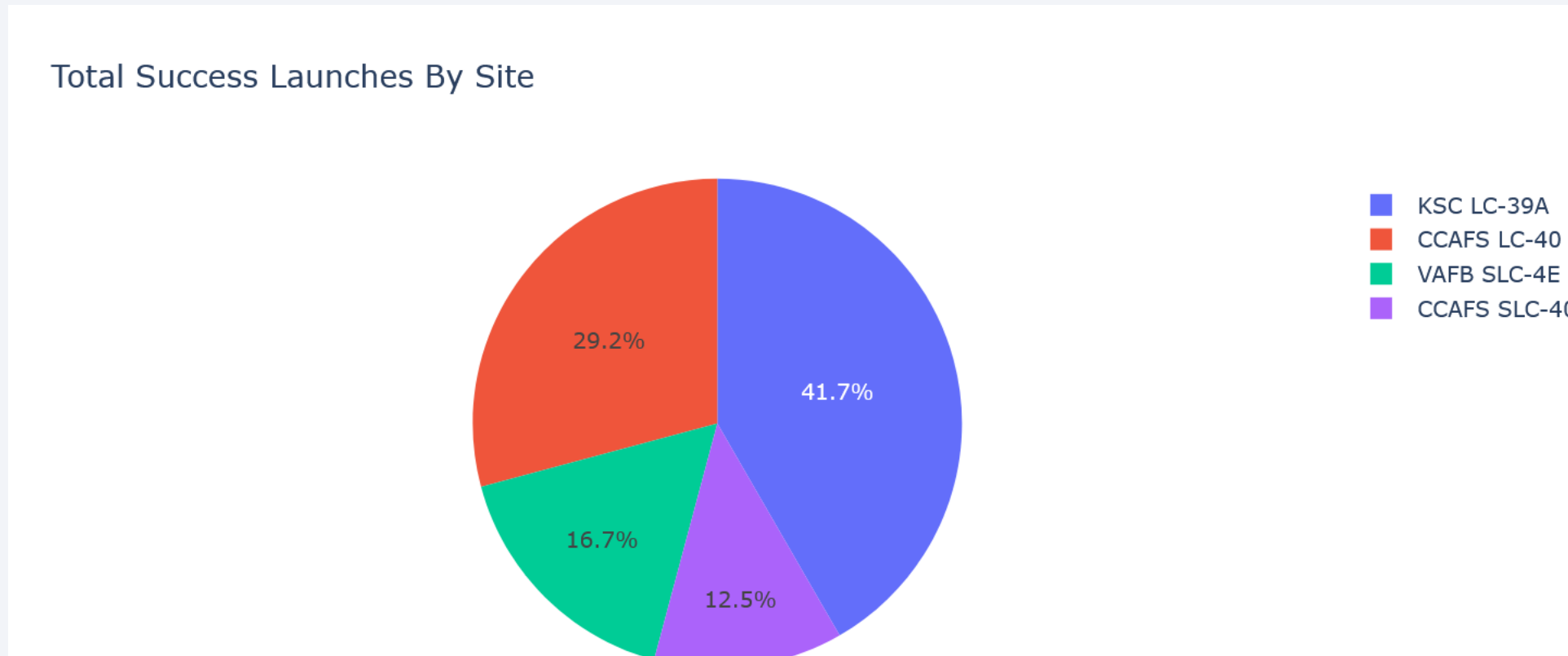


Section 4

Build a Dashboard with Plotly Dash

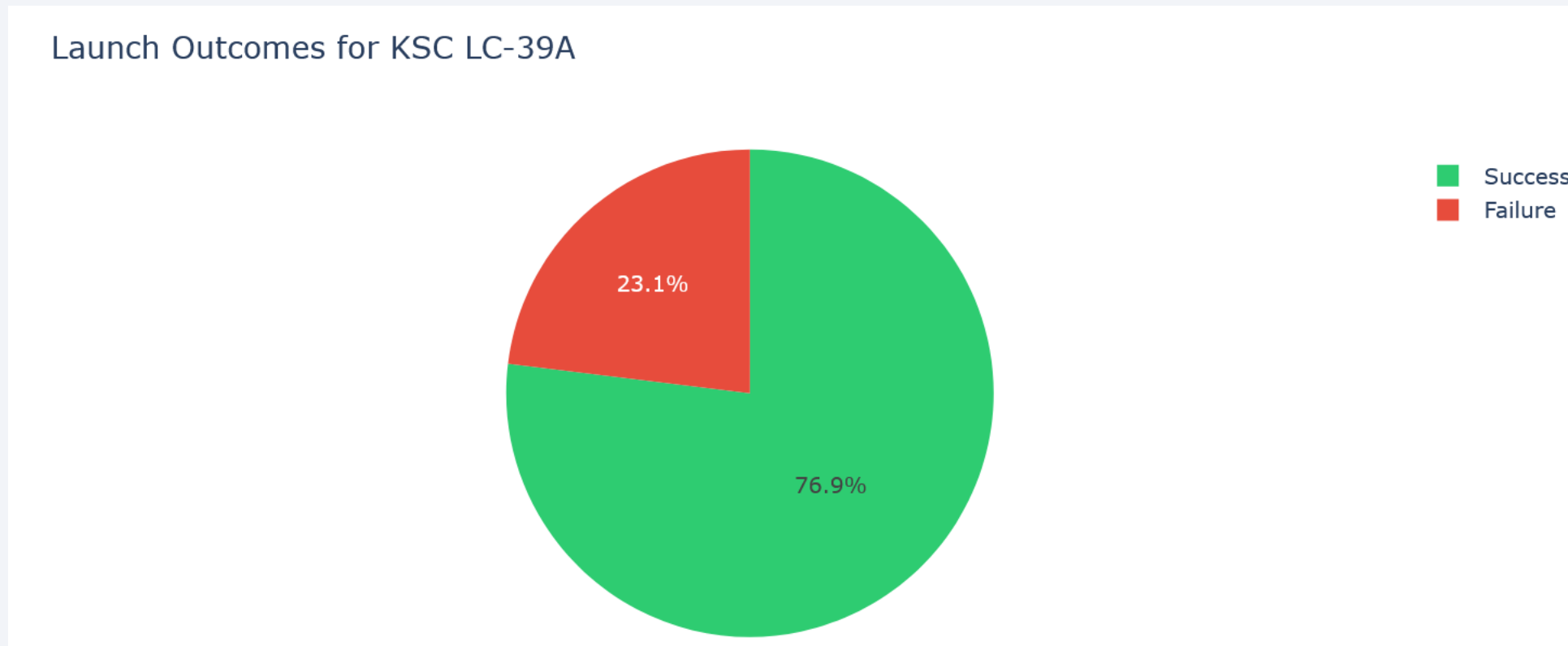
Total Success Launches for site

- Explain the important elements and findings on the screenshot



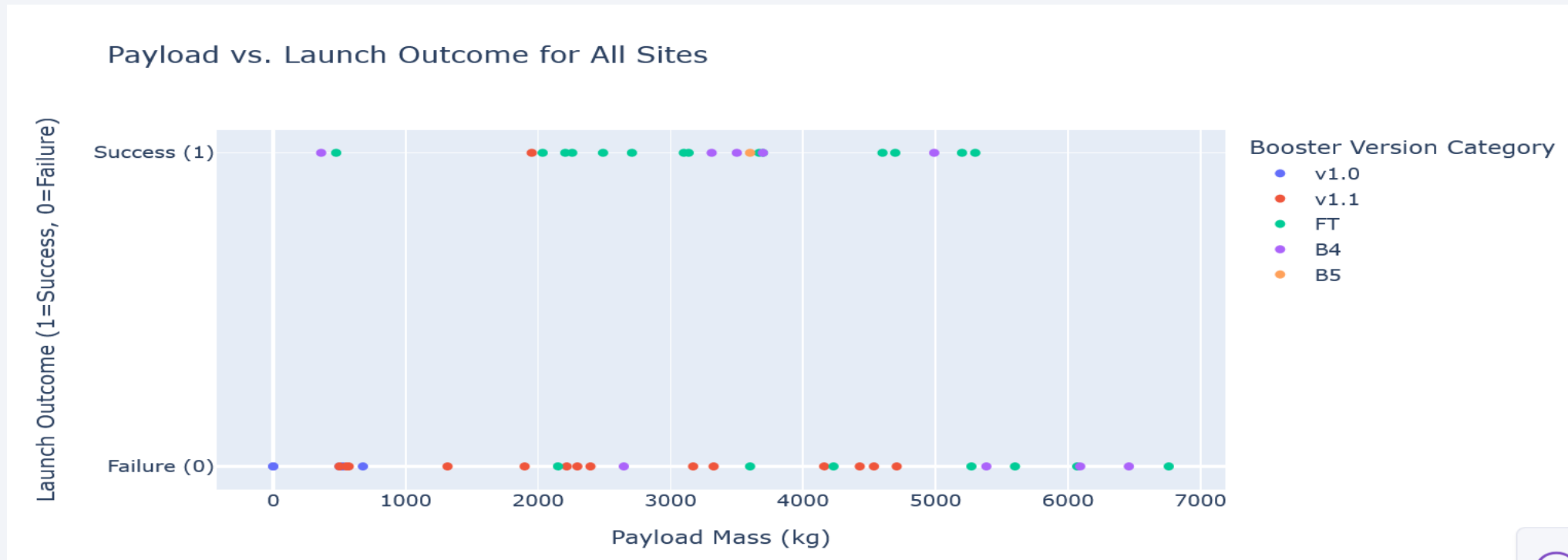
Launch site with highest launch success ratio

- Explain the important elements and findings on the screenshot



Payload vs. Launch Outcome scatter plot for all sites

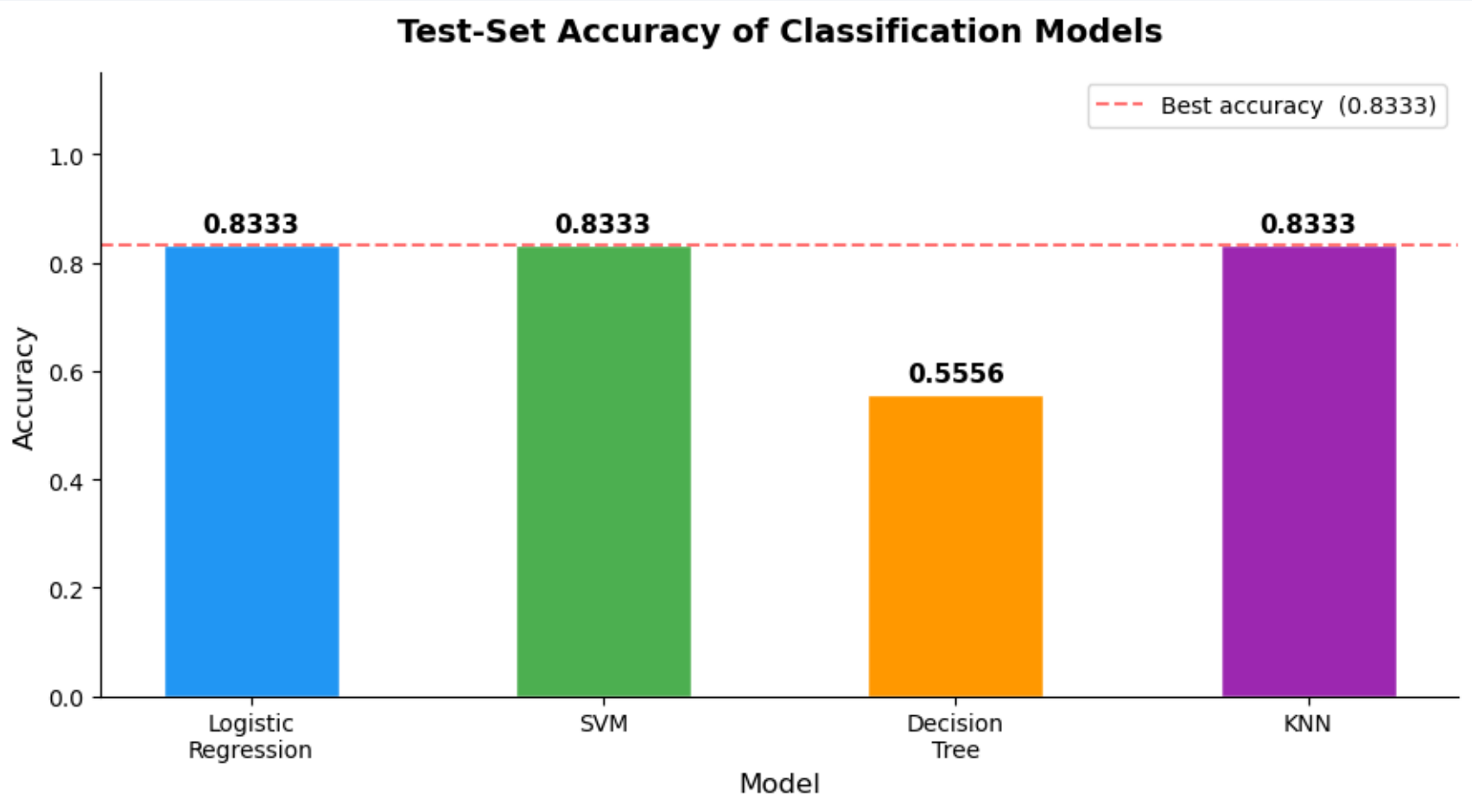
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.



Section 5

Predictive Analysis (Classification)

Classification Accuracy



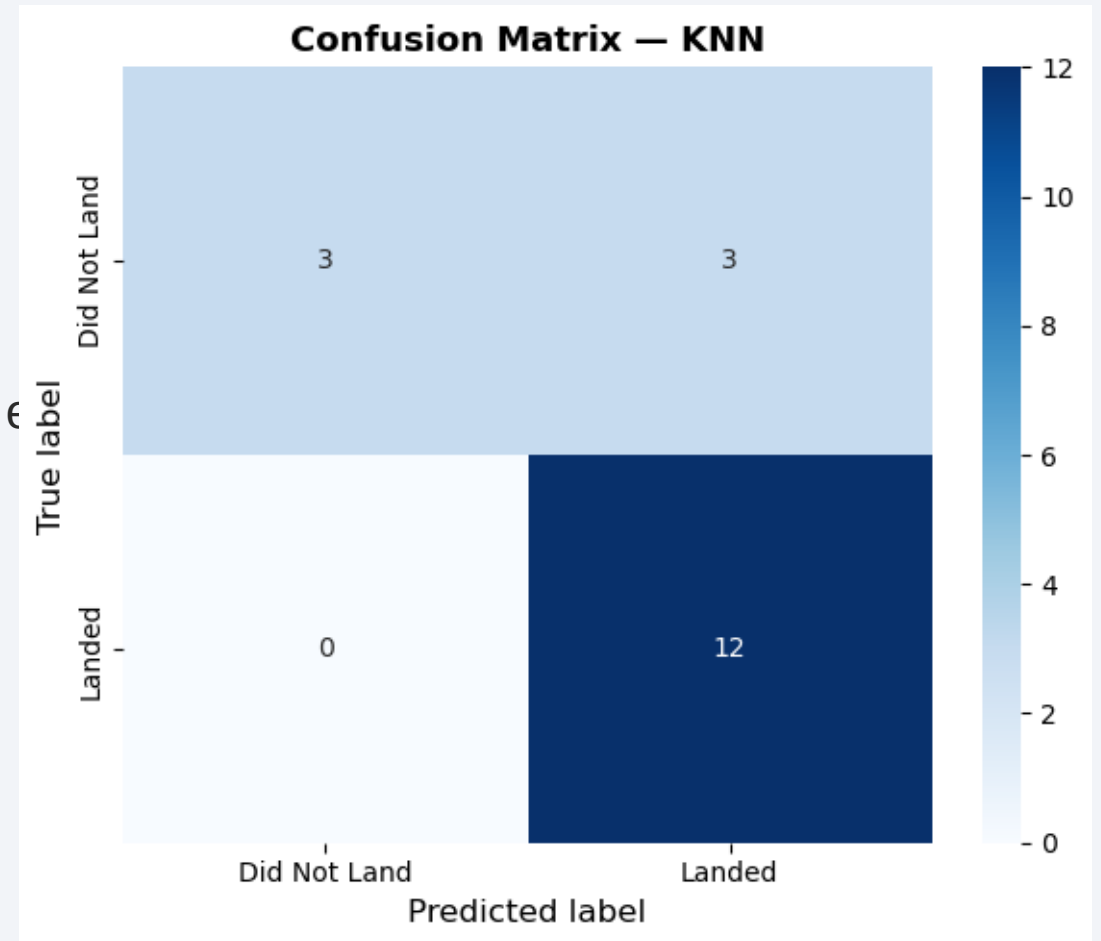
Confusion Matrix

Best Model: KNN

Test Accuracy: 0.8333

Validation: 0.8482

Key takeaway: In the business context of bidding against SpaceX, False Positives are the more costly error over-bid assuming an expensive launch when SpaceX actually recovers the booster cheaply. Conversely, False Negatives are missed chances to under-bid strategically. Minimising FP (high Precision on the 'Landed' class) should be prioritised alongside overall accuracy.



Conclusions

- **Launch Success Trends:** The yearly trend chart shows that SpaceX's launch success rate improved consistently from 2010 through 2019, demonstrating significant operational maturity over time.
- **Best Launch Site:** KSC LC-39A had the highest success ratio among all sites at 76.9%, and also accounted for the largest share (41.7%) of total successful launches.
- **Payload & Orbits:** The majority of payloads were under 7,000 kg, and four orbit types were equally represented. ES-L1, GEO, and HEO orbits achieved 100% success rates.
- **Mission Outcomes:** Out of 101 recorded missions, 98 were successful, with only 1 in-flight failure — a very strong overall success record.
- **Predictive Modeling:** The KNN classifier performed best among the four models tested (Logistic Regression, SVM, Decision Tree, KNN), achieving a test accuracy of 83.33% and validation accuracy of 84.82% in predicting first-stage landing success.
- **Core Business Insight:** By predicting whether SpaceX will successfully recover the Falcon 9 first stage, a competitor like "Space Y" can better estimate launch costs, since reusability is the primary driver of SpaceX's cost advantage.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

